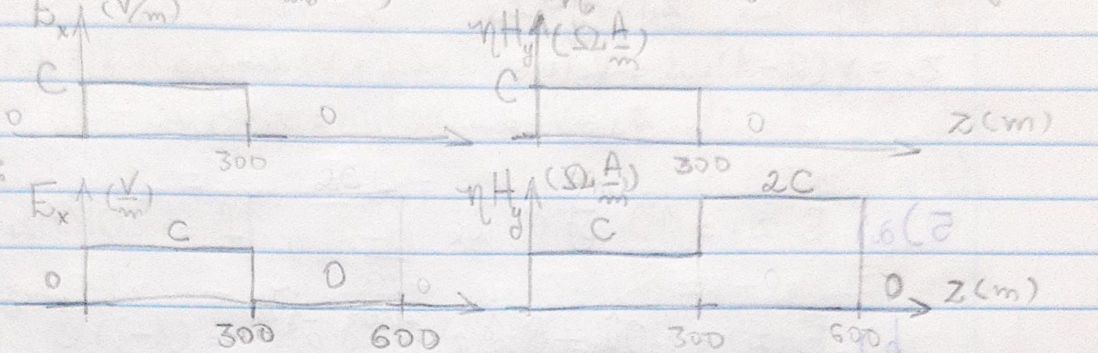
LISTA 3

1) $z_1 = 3.10^8 . 10^{-6} = 300\,m$ $\quad z_2 = 3.10^8 . 3.10^{-6} = 900\,m$

$E_x^+ = C\,u(t) = -E_x^-$ $(\sigma_{cond} \to \infty)$ $\quad H_y^+ = \frac{E_x^+}{\eta_0} = C/\eta \cdot u(t) = +H_y^-$

$t_1 = 1\mu s$:

$t_2 = 3\mu s$:



2) a) Na superfície: $\vec{J}_s = \underbrace{\hat{u}_n}_{\hat{u}_z} \times (\underbrace{\vec{H}_{y2}}_{0} - \underbrace{\vec{H}_{y1}}_{2C/\eta\ \hat{u}_y})$

$\underbrace{-(\hat{u}_z \times \hat{u}_y)}_{-\hat{u}_x} \frac{2C}{\eta} \to \vec{J}_s = +\frac{2C}{\eta}\,\hat{u}_x$

$P_f = \frac{1}{2} |J|^2 \eta = \frac{2C^2}{\eta}$

b) $t = 1\mu s \to$ ainda não há onda refletida e $P_f = C^2/\eta$ na região $200 < z < 400\,m$

c) $z < 300$: $w_E = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{\epsilon C^2}{2}$; $w_H = \frac{1}{2} \vec{H} \cdot \vec{B} = \frac{\mu C^2}{\eta^2} = w_E$

$\to$ igualmente distribuída nos campos

$z > 300$: $w_E = 0$ e $w_H = 2\epsilon C^2$

A potência em $z = 400$ é nula, armazenada como campo magnético.

3) $f = 1\,MHz$ $\quad \vec{E} = (4\,\hat{u}_x + 8 e^{j\frac{\pi}{3}}\,\hat{u}_y + 4\sqrt{3}\, e^{j\frac{\pi}{2}}\,\hat{u}_z) \frac{V}{m}$

$\vec{E} = 4 . \cos(\omega t)\,\hat{u}_x + 8 . \cos(\omega t + \pi/3)\,\hat{u}_y - 4\sqrt{3}\,\mathrm{sen}(\omega t)\,\hat{u}_z$

$= 4 \cos(\omega t)\,\hat{u}_x + (4 \cos(\omega t) - 4\sqrt{3}\,\mathrm{sen}(\omega t))\,\hat{u}_y - 4\sqrt{3}\,\mathrm{sen}(\omega t)\,\hat{u}_z$

Numa base ortogonal (cosseno e seno) c/ normal ($\hat{n} = c \times s$):

$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & 4 & 0 \\ 0 & -4\sqrt{3} & -4\sqrt{3} \end{vmatrix} = 16\sqrt{3} \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$, contida no plano $-x + y - z = 0$,

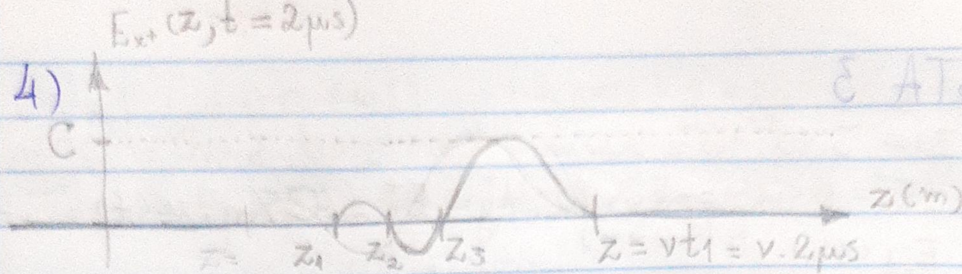
ou $z = y - x$

$|\vec{E}| = \sqrt{16 \cos^2(\omega t) + 64 \cos^2(\omega t + \pi/3) + 48\,\mathrm{sen}^2(\omega t)} \to 4{,}6549 < |\vec{E}| < 10{,}3117 \frac{V}{m}$

$= \sqrt{32} \cdot \sqrt{1 + 2\,\mathrm{sen}^2(\omega t) - \sqrt{3} \cos(\omega t)\,\mathrm{sen}(\omega t)}$

②

4)



$z_1 = v(2-1)\mu s = v.1\mu s$ $\quad z_2 = v(2-0{,}8)\mu s = v.1{,}2\mu s$

$z_3 = v(2-0{,}6)\mu s = v.1{,}4\mu s$

5) a) $\dot{\vec{E}}(z) = \frac{10}{\sqrt{2}} e^{-j2z}\, \hat{u}_x \ \frac{V_{ef}}{m}$, $f = 100MHz$

b) $\dot{\vec{E}}(z) = \left(\frac{10\hat{u}_x}{\sqrt{2}} + \frac{8}{\sqrt{2}} e^{j\pi} \hat{u}_y\right) e^{+j30z} \underbrace{e^{-j\pi/4}}_{\frac{1}{\sqrt{2}}(1-j)} = (1-j)\, e^{j30z} (5\hat{u}_x - 4\hat{u}_y)$ $\left[V_{ef}/m\right]$

$f = 1GHz$

c) $\dot{\vec{E}}(z) = \frac{(4\hat{u}_x + 3e^{j\pi}\hat{u}_z)}{\sqrt{2}} e^{-j2z}\, e^{j5x}\, e^{j\pi/8} \ \frac{V_{ef}}{m}$, $f = 1GHz$

d) $\dot{\vec{E}}(z) = \frac{e^{-j\pi}}{\sqrt{2}} (5\hat{u}_x + 10\hat{u}_y) e^{-2z}\ e^{-j2z}\ e^{j\pi/8}$

$\dot{\vec{E}}(z) = \frac{1}{\sqrt{2}} (5\hat{u}_x + 10\hat{u}_y) e^{-2z(1+j)}\ e^{-j\frac{7\pi}{8}} \ \frac{V_{ef}}{m}$, $f = 100MHz$

e) $\dot{\vec{E}}(z) = \frac{3e^{-2z}\hat{u}_x}{\sqrt{2}} . e^{-j2z} . e^{j\pi/4} = (1{,}5 + j1{,}5)\, e^{-2z(1+j)} \frac{V_{ef}}{m}$

$f = 100MHz$

6) Caso 1: $jk = (1+j)\sqrt{\pi f \mu \sigma} = (1+j)\, 1560{,}71 = \alpha + j\beta$

$\dot{E} = 1\angle 0\ V_{ef}/m$ $\quad \eta = \sqrt{\frac{2\pi f \mu}{\sigma}} \angle 45 = 35{,}77 \angle 45°\ \mu\Omega$

a) $\dot{H} = \frac{\hat{u}_z \times \dot{\vec{E}}}{\eta} = 27954{,}24 \angle -45°\ A_{ef}/m$

b) $\measuredangle H = -45° = -\pi/4$ rad

c) $N = Re\{E \times H^*\} = Re\left\{\frac{|E|^2}{\eta^*}\right\} = 19766{,}634\ W/m^2$

d) $\lambda_0 = c/f = 30Km$ $\quad \dot{E} \approx 0 . e^{+j2{,}083} = 0$ $\quad \dot{H} \approx 0 . e^{j1{,}2975} = 0$

$N = 0$

Caso 2: $jk = (1+j)\,49354$ 1/m; $\eta = (1+j)\sqrt{\dfrac{\mu f \pi}{\sigma}} = 1{,}13\angle 45°$ mΩ

a) $|\dot{H}| = 883{,}991$ Aef/m b) $-45° = -\pi/4$

c) $N = 625{,}076$ W/m² d) $\lambda_o = 30$ m $\dot{E} \approx 0 \cdot e^{j0{,}0256} = 0$

Caso 3: $jk = (1+j)\,527{,}94$ $\eta = 105{,}753\angle 45°$ $\mu\Omega$

a) $|\dot{H}| = 9456$ Aef/m b) $-45° = -\pi/4$

c) $N = 6686{,}4$ W/m² d) $\dot{E} \approx 0 \cdot e^{j0{,}5665} = 0$

Caso 4: $jk = (1+j)\,16694{,}84$ $\eta = 3{,}344\angle 45°$ mΩ

a) $|\dot{H}| = 299{,}025$ Aef/m b) $-45° = -\pi/4$

c) $N = 211{,}44$ W/m² d) $\dot{E} \approx 0 \cdot e^{j0{,}0336} = 0$

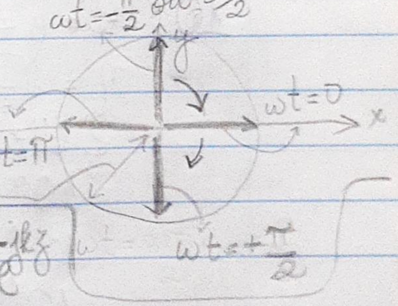
7) a) $\vec{E}(z,t) = \sqrt{2}\,\dot{E}_{xo+}\, e^{-\alpha z}\left[\cos(\omega t - \beta z)\,\hat{u}_x - \operatorname{sen}(\omega t - \beta z)\,\hat{u}_y\right]$

$z=0,\ t=0:\ \vec{E} = \sqrt{2}\,\dot{E}_{xo+}\,(\cos(0)\,\hat{u}_x + \cos(\pi/2)\,\hat{u}_y)$

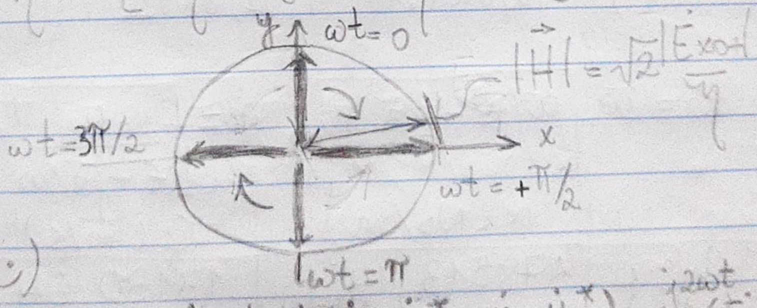
$z=0,\ \omega t=\pi/2:\ \vec{E} = \sqrt{2}\,\dot{E}_{xo+}\,(0\,\hat{u}_x - 1\,\hat{u}_y)$

$\omega t = \pi:\ \vec{E} = -\sqrt{2}\,\dot{E}_{xo+}\,(+\hat{u}_x)$

$\omega t = -\pi/2:\ \vec{E} = +\sqrt{2}\,\dot{E}_{xo+}\,(\hat{u}_y)$



b) $\dot{H} = \dfrac{\hat{u}_z \times \dot{E}}{\eta} = \left[\dfrac{\dot{E}_{xo+}}{\eta}\,\hat{u}_y + j\,\dfrac{\dot{E}_{xo+}}{\eta}\,(-\hat{u}_x)\right] e^{-jkz}$



(circular à esquerda :))

c) $\vec{N}(z=0,t) = \hat{u}_z\,\operatorname{Re}\{(\dot{E}_x\dot{H}_y^* - \dot{E}_y\dot{H}_x^*) + e^{j2\omega t}(\dot{E}_x\dot{H}_y - \dot{E}_y\dot{H}_x)\}$

$= \operatorname{Re}\left\{\dfrac{|\dot{E}_{xo+}|^2}{\eta} - j^2\dfrac{|\dot{E}_{xo+}|}{\eta}\right\} + \operatorname{Re}\left\{e^{j2\omega t}\cdot 2\,\dfrac{|\dot{E}_{xo+}|}{\eta}\right\}\hat{u}_z$

$= 2\cdot\dfrac{|\dot{E}_{xo+}|^2}{\eta}\left[1 + \cos(2\omega t)\right]\hat{u}_z$

d) $\vec{E}(z=0,t) = E_0\cos(\omega t)\,\hat{u}_x$

$\vec{H}(z=0,t) = E_0/\eta\,\cos(\omega t)\,\hat{u}_y$ $\vec{N}(z=0,t) = \dfrac{E_0^2}{\eta}\left[1+\cos(2\omega t)\right]\hat{u}_z$

4

8. a) $\varphi_y - \varphi_x = +\pi/2 \Rightarrow$ circular à esquerda

b) vácuo $\Rightarrow \eta = \eta_0 = 377\ \Omega$

$\vec{E}_+(z,t) = \sqrt{2}\,E_0 \cos(\omega t - \beta z)\,\hat{u}_x - \sqrt{2}\,E_0\, sen(\omega t - \beta z)\,\hat{u}_y$

$= 2\sqrt{2}\cos(\omega t - \beta z)\,\hat{u}_x - 2\sqrt{2}\, sen(\omega t - \beta z)\,\hat{u}_y$ mV/m

$\vec{H}_+(z,t) = 5{,}3\sqrt{2}\, sen(\omega t - \beta z)\,\hat{u}_x + 5{,}3\sqrt{2}\cos(\omega t - \beta z)\,\hat{u}_y$ µA/m

c) $\vec{N}_{med} = Re\{\dot{E}\times\dot{H}^*\} = 2\dfrac{E_0^2}{\eta}\hat{u}_z \cong 21{,}2\ \hat{u}_z$ nW/m²

9. $\vec{E}_+(z,t) = 0{,}2\sqrt{2}\cos(\omega t - \beta z)\,\hat{u}_x - 0{,}2\sqrt{2}\, sen(\omega t - \beta z)\,\hat{u}_y$ V/m

$\eta = \eta_0/\sqrt{\epsilon_r} = 188{,}50\ \Omega$

$\vec{H}_+(z,t) = 1{,}061\sqrt{2}\,[sen(\omega t - \beta z)\,\hat{u}_x + \cos(\omega t - \beta z)\,\hat{u}_y]$ mA/m

10. $k = \beta - j\alpha = \omega\sqrt{\mu(\epsilon' - j\epsilon'')} \approx \omega\sqrt{\mu\epsilon'}\left(1 - j\dfrac{\epsilon''}{2\epsilon'}\right)$

$\eta = \sqrt{\dfrac{\mu}{\epsilon' - j\epsilon''}} \approx \sqrt{\mu/\epsilon'}$ $\quad v_f = \omega/\beta$

f (Hz)	α (Np/m)	v_f (10^8 m/s)	η (Ω)
10^6	$4{,}585\cdot 10^{-7}$	1,8737	235,46
10^8	$6{,}562\cdot 10^{-5}$	1,8774	235,92
10^{10}	$2{,}827\cdot 10^{-2}$	1,8811	236,38

11. $k = \omega\sqrt{\mu\left(\epsilon - j\dfrac{\sigma}{\omega}\right)}$; $\eta = \sqrt{\dfrac{\mu}{\epsilon - j\sigma/\omega}}$ e $v_f = \dfrac{\omega}{\beta}$

	α (Np/m)	v_f (m/s)	η (Ω)
Cobre	478513,14	13130,64	$0{,}0117\angle 45°$
Água	77,45	30814449,58	$36{,}2\angle 20{,}8°$

12. a) $\eta = (188{,}36\angle 0{,}23°)\,\Omega$; $k = (2{,}096 - j0{,}0105)$ rad/m

$\dot{E}_x(z) = 10\, e^{-j2{,}096z} e^{-0{,}0105z}\,\hat{u}_x$ mVef/m

$\dot{H}_y(z) = 53{,}09\, e^{-j(2{,}096z - 0{,}005)} e^{-0{,}0105z}\,\hat{u}_y$ µAef/m

$E_x(z,t) = \sqrt{2}\,10\, e^{-0{,}0105z}\cos(\omega t - 2{,}096z)\,\hat{u}_x$ mV/m

$H_y(z,t) = \sqrt{2}\,53{,}09\, e^{-0{,}0105z}\cos(\omega t - 2{,}096z)\,\hat{u}_y$ µA/m

b) $\vec{N}_{méd}(z) = \frac{|E_{xo+}|^2}{\eta} e^{-2\alpha z} \hat{u}_z = 0{,}5309\, e^{-0{,}021 z}\, \hat{u}_z\ \mu W/m^2$

c) $\Delta\vec{N} = \underbrace{N_o\, e^{-2\alpha z_1}}_{N_{med}(z_1)} (e^{-2\alpha \Delta z} - 1)\, \hat{u}_z\ \mu W/m$

$\approx N_{méd}(z_1)\, 2\alpha \Delta z\ \mu W/m^2$

$P_{diss} = \int_V \omega\epsilon'' . |E_x|^2 dV = \omega\epsilon'' \underset{1}{A} \Delta z\, |E_{xo+}|^2 e^{-2\alpha z_1}$

Como $\frac{2\alpha}{\eta} = \frac{0{,}021}{188{,}5} = 1{,}114 . 10^{-4}\ \frac{Np}{m.\Omega}$

e $\omega\epsilon'' = 2\pi . 50 . 10^6 . 0{,}04 . 8{,}854 . 10^{-12} = 1{,}113 . 10^{-4}\ \frac{S}{m}$

~~são~~ aproximadamente iguais, conclui-se que $\Delta\vec{N} \approx P_{diss}$

13. a) $\delta = (\pi f \mu \sigma)^{-1/2} = 66{,}085\ \mu m$

b) $\eta = (368{,}96 \angle 45°)\ \mu\Omega$; $k = 15131{,}914\,(1 - j)$

$\vec{H}(z) = 2\, e^{-j 15132 z}\, e^{-15132 z}\, \hat{u}_y\ Aef/m$

$\vec{E}(z) = 0{,}738\, e^{-j 15132 z}\, e^{-15132 z}\, e^{j\pi/4}\, \hat{u}_x\ mVef/m$

$\vec{J}(z) = 42{,}8\, e^{-j(15132 z + \pi/4)}\, e^{-15132 z}\, \hat{u}_x\ kAef/m^2$

c) i) $\vec{N}(z=0) = Re\{|H|^2 . \eta\} = 1{,}0436\, \hat{u}_z\ \frac{mW}{m^2}$

ii) $\int_0^\infty |J|^2/\sigma\, dz = \frac{\delta}{2} 31{,}582\, \hat{u}_z\ \frac{W}{m^2}$

14. a) $\eta = \eta_0$; $k = 20{,}958$ rad/m

$\vec{E}(z,t) = 5\sqrt{2} \cos(\omega t - 20{,}958 z)\, \hat{u}_x + 5\sqrt{2}\, sen(\omega t - 20{,}958 z)\, \hat{u}_y\ V/m$

b) $\vec{H}(z,t) = 13{,}26\sqrt{2} [\cos(\omega t - 20{,}958 z)\hat{u}_y + - sen(\omega t - 20{,}958 z)\hat{u}_x]\ \frac{mA}{m}$

c) $\varphi_y - \varphi_x = -\pi/2$; $|Ex| = |Ey| \Rightarrow$ circular à direita.

d) $\vec{N}_{med}(z) = Re\{\vec{E} \times \vec{H}^*\} = (E_x H_y^* - E_y H_x^*)\, \hat{u}_z$

$= (|E_0|^2/\eta \overset{+}{-} j^2 |E_0|^2/\eta)\hat{u}_z = 2 . \frac{25}{377} \hat{u}_z = 132{,}626\, \hat{u}_z\ m\frac{W}{m^2}$

15. a) $\gamma = jk = j2\pi\, 200 \cdot 10^6 \sqrt{\mu_0 \epsilon_0} = j4,189$ Np/m, com $\sqrt{\mu_0 \epsilon_0} = 1/c$

$\eta = \eta_0 = 377\,\Omega$

b) $\vec{E}_+(z) = 3 e^{-\gamma z}\,\hat{u}_x + j3 e^{-\gamma z}\,\hat{u}_y$ Vef/m

$\vec{H}_+(z) = -j8 e^{-\gamma z}\,\hat{u}_x + 8 e^{-\gamma z}\,\hat{u}_y$ mAef/m

c) $\vec{E}_+(z,t) = 3\sqrt{2}\cos(2\pi f t - kz)\,\hat{u}_x$

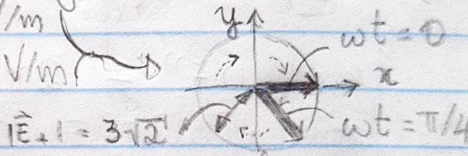
$- 3\sqrt{2}\,\text{sen}(2\pi f t - kz)\,\hat{u}_y$ V/m

d) $\vec{E}_+(z=0,t) = 3\sqrt{2}\left(\cos(\omega t)\,\hat{u}_x + \text{sen}(\omega t)\,\hat{u}_y\right) \frac{V}{m}$

Que se "distribui" nos versores ao longo de t,

e vale: $\vec{E}_+(z=0, t=0) = 3\sqrt{2}\,\hat{u}_x$ V/m

já $\vec{E}_+(z=0, \omega t=\pi/4) = 3(\hat{u}_x - \hat{u}_y)$ V/m

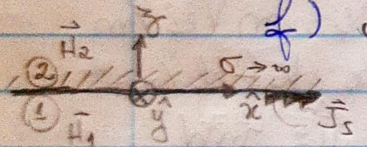


$\varphi_y - \varphi_x = +\pi/2$ e $|E_x| = |E_y|$ ⇒ polarização circular à esquerda.

e) Campo elétrico não propaga em condutor ($\rho = -1$)

⇒ $\vec{E}_-(z) = -3 e^{+\gamma z}[\hat{u}_x + j\hat{u}_y]$ Vef/m

$\vec{H}_-(z) = 8 e^{+\gamma z}[-j\hat{u}_x + \hat{u}_y]$ mAef/m



f) $\vec{J}_s = \hat{u}_z \times (\vec{H}_2 - \vec{H}_1) = (\vec{H}_{1+}(z=0) + \vec{H}_{1-}(z=0)) \times \hat{u}_z$

$= 16\,[j\hat{u}_y + \hat{u}_x]$ mAef/m

16. a) $\gamma = j2\pi\, 10^{10}/c = j209,44$ Np/m

$\lambda = c/f = 3\cdot 10^8/10^{10} = 3$ cm

b) Circular à esquerda.

c) $\eta_2 = \sqrt{\dfrac{\mu_0}{\epsilon_0 - j\sigma/\omega}} = 139,825\angle 21^\circ$; $\rho_0 = 0,583\angle 150,52^\circ$

$\vec{E}_-(z) = \rho_0 E_0\, e^{+\gamma z}\,\hat{u}_x + j\rho_0 E_0\, e^{+\gamma z}\,\hat{u}_y$

$= 1,167\exp\{j(209,44z + 2,627)\}\cdot[\hat{u}_x + j\hat{u}_y]$ $\frac{mVef}{m}$

$\vec{E}_-(z,t) = 1,167\sqrt{2}\,[\cos(20\pi\, 10^9 t + 209,44z + 2,627)\,\hat{u}_x$

$- \text{sen}(20\pi\, 10^9 t + 209,44z + 2,627)\,\hat{u}_y]$ $\frac{mV}{m}$

d) A onda refletida propaga-se no sentido de $-\hat{u}_z$, e nessa referência ⇒ polarização circular à direita.

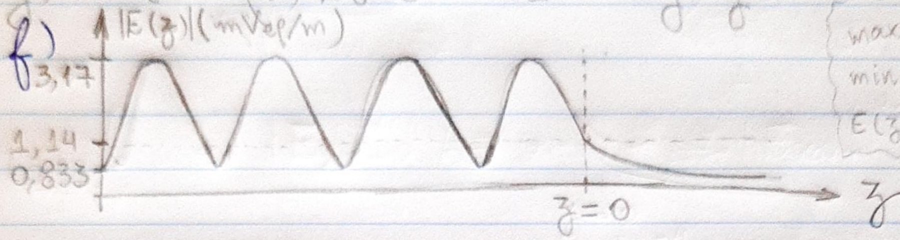
Transmitida permanece à esquerda.

e) $z<0$ (vácuo): $\vec{E}_1(z) = E_0 e^{-\gamma z}(1+\rho_0 e^{+2\gamma z})[\hat{u}_x + j\hat{u}_y]$

$z>0$ (condutor): $\gamma_2 = j2\pi 10^{10}\sqrt{\mu_0(\epsilon_0 - j2/\pi 10^{10})} = 370{,}77 + j425{,}91$

$\vec{E}_2(z) = (1+\rho_0) E_0 e^{-\gamma_2 z}[\hat{u}_x + j\hat{u}_y]$

f)



$\max(|E_1|) = (1+|\rho|)E_0$

$\min(|E_1|) = |E_{max}|/COE$

$E(z=0) = |1+\rho|\cdot E_0$

17. a) $\eta_1 = \eta_0/\sqrt{\epsilon_r} = \eta_0/2$; $\rho_0 = -\frac{1/2}{3/2} = -1/3$

$\vec{N}_0 = N_+(1-|\rho_0|^2)\hat{u}_z = 4{,}716\,\hat{u}_z$ mW/m²

$\vec{N}_1 = 4{,}716\,\hat{u}_z$ mW/m²

b) $k_0 = 2\pi 3.10^9/c = 20\pi$; $k_1 = 40\pi$ rad/m

$\dot{E}_0(-0{,}14) = E_+ e^{-jk_0(-0{,}14)}(1+\rho_0 \cdot e^{-j2k_0 0{,}14}) = 1{,}345\angle 124{,}54°$ Vef/m

$\dot{H}_0(-0{,}14) = E_+/\eta_0\, e^{jk_0 0{,}14}(1-\rho_0 e^{-j2k_0 0{,}14}) = 4{,}305\angle 160{,}03°$ mAef/m

$\dot{E}_1(0{,}14) = (1+\rho_0)E_+ e^{-jk_1 0{,}14} = 0{,}943\angle 72°$ Vef/m

$\dot{H}_1(0{,}14) = (1+\rho_0)E_+/\eta_2\, e^{-jk_1 0{,}14} = 5\angle 72°$ mAef/m

c) $\rho(-0{,}025) = \rho_0 \cdot \exp\{-j2k_0 0{,}025\} = -\rho_0 = 1/3$

$Z(-0{,}025) = 754\,\Omega$; $Z(0{,}025) = \eta_1 = 188{,}5\,\Omega$

18. a) $k_1 = 20\pi\sqrt{4-j0{,}04} = \pi(40-j0{,}2)\ \frac{rad}{m} = \beta_1 - j\alpha_1$

$\eta_1 = \eta_0/\sqrt{4-j0{,}04} = 188{,}5\angle 0{,}286°\,\Omega$; $\rho_0 = 1/3\angle 179{,}62°$

$\vec{N}_0 = 4{,}716\,\hat{u}_z$ mW/m² ; $\vec{N}_1(z) = 4{,}716\, e^{-\pi 0{,}4 z}\cdot\hat{u}_z$ mW/m²

b) $\dot{E}_0(-0{,}14) = 1{,}342\angle 124{,}53°$ Vef/m

$\dot{H}_0(-0{,}14) = 0{,}004312\angle 159{,}974°$ Aef/m

$\dot{E}_1(0{,}14) = 0{,}863\angle 72{,}2°$ Vef/m

$\dot{H}_1(0{,}14) = 0{,}00458\angle 71{,}9°$ Aef/m

c) $\rho(-0{,}025) = 1/3\angle -0{,}382°$

$Z(-0{,}025) = 754{,}019\angle -0{,}286°\,\Omega$

$Z(0{,}025) = 188{,}4953\angle 0{,}286°\,\Omega$

19. A frequência efetiva que incide no plano é $f' = f(1 + v/c)$ (efeito Doppler).
Igualmente para a onda refletida $f'' = f'(1 + v/c)$ (efetivamente emitida do plano, com velocidade v)

$$f'' = f(1+v/c)^2 = f(1 + 2v/c + v^2/c^2) \approx f(1+2v/c)$$

$$\Delta f = f'' - f = 2fv/c = 2 \cdot 10^{10} \cdot 100/1{,}08 \cdot 10^9 = 1853\,Hz$$

20. a) $|\rho| = \dfrac{COE - 1}{COE + 1} = 1/2$; $\rho_{min} = |\rho| \angle 180° = -1/2$

$\beta_0 = k_0 = 2\pi \, 300 \cdot 10^6 / c$ $\qquad \rho(z=0) = \rho_{min} \cdot e^{+2j\beta_0 \cdot 0{,}5} = \rho_{min}$

b) $\eta_{diel} = \dfrac{1 + \rho(z=0)}{1 - \rho(z=0)} \eta_0 = \dfrac{\eta_0}{COE} = 125{,}67\,\Omega$

c) Para um dielétrico de pequenas perdas, vale $\epsilon'' \ll \epsilon'$, equivalente a $\sigma \approx 0$, como pode ser observado em η_{diel} – aproximado como real –, e na atenuação $\alpha = 0{,}001 \ll 1$.

d) $\eta_{diel} \approx \sqrt{\dfrac{\mu_0}{\epsilon'}} \Rightarrow \epsilon' = 9\epsilon_0$; $\alpha \approx \omega \sqrt{\dfrac{\mu_0}{\epsilon'}} \cdot \dfrac{\epsilon''}{2} \Rightarrow \dfrac{\epsilon''}{\epsilon_0} = \dfrac{2}{\epsilon_0} \cdot \dfrac{\alpha}{\omega \cdot \eta_{diel}}$

$v_f = \omega/\beta = 10^8\ m/s$ $\qquad \epsilon = (9 - j0{,}00095)\epsilon_0$

e) $E_{x0-} = \rho(z=0) E_{x0+} = -1\ V_{ef}/m$

$E_{x0,t} = (1 + \rho(z=0)) E_{x0+} = +1\ V_{ef}/m$

f) $\vec{N}_{ar}(z) = \dfrac{|E_{x+}|^2}{\eta_0}(1 - |\rho|^2) = \dfrac{|E_{x0,t}|^2}{\eta_{diel}} = 7{,}958\ \hat{u}_z\ mW/m^2$

$\vec{N}_{diel}(z) = 7{,}958 \cdot e^{-0{,}002z}\ \hat{u}_z\ mW/m^2$

21. a) $v = \omega/\beta = 400 \cdot 10^6/2 = 2 \cdot 10^8\ m/s$

$\lambda = 2\pi/\beta = 50\,cm$

Para maiores perdas, o módulo de η_2 diminuiria, e sua fase aproximaria-se de 45°.

b) $\rho(z=0) = 0{,}2018 \angle 172{,}84° \approx -0{,}2$

$\dot{E}^-_{x1}(z=0) = 0,605 \angle 172,84°$ mVef/m

$\dot{E}_{x2}(z=0) = 2,401 \angle 1,80°$ mVef/m

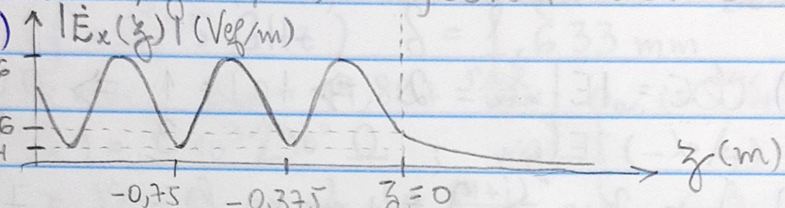
c) $\dot{E}_1(z) = \left(3\, e^{-j\frac{2\pi}{0,75}\cdot z} + 0,605\cdot e^{j(3,017+\frac{2\pi}{0,75}z)}\right)\hat{u}_x$ mVef/m

$\dot{H}_1(z) = \left(7,96\, e^{-j8,378z} - 1,606\, e^{j(3,017+8,378z)}\right)\hat{u}_y$ μAef/m

$\dot{E}_2(z) = 2,401\, e^{-j(4\pi z - 0,0314)}\, e^{-0,63\cdot z}\,\hat{u}_x$ mVef/m

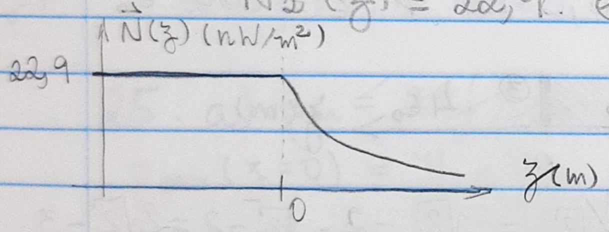
$\dot{H}_2(z) = 14,331\, e^{-j(4\pi z + 0,0733)}\, e^{-0,63z}\,\hat{u}_y$ μAef/m

d) $E_x(t, z=1,25) = 0,001545\cos(\omega t - 3,1017)\,\hat{u}_x$ V/m

e) 

f) $\vec{N}_1(z) = 22,9\,\hat{u}_z$ nW/m²

$\vec{N}_2(z) = 22,9\cdot e^{-1,26z}\,\hat{u}_z$ nW/m²



No ar, não há gerador e a potência é constante. No dielétrico, a potência sofre perdas e aquece o mesmo.

22. a) $\eta_{diel} = \eta_0/\sqrt{\epsilon_r} = 125,67\,\Omega$; $\rho(z=0) = -1/2$

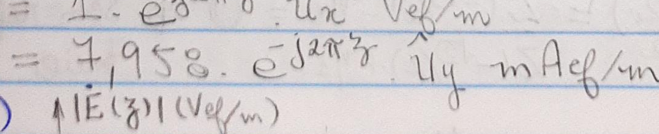
b) $k_0 = \beta_0 = 2\pi f/c = \frac{2}{3}\pi$; $k_{diel} = \beta_0\cdot\sqrt{\epsilon_r} = 2\pi$ rad/m

$\dot{E}_0(z) = \left(2\, e^{-j\frac{2}{3}\pi z} - 1\, e^{j\frac{2}{3}\pi z}\right)\hat{u}_x$ Vef/m

$\dot{H}_0(z) = \left(5,3\, e^{-j\frac{2}{3}\pi z} + 2,65\, e^{j\frac{2}{3}\pi z}\right)\hat{u}_y$ mAef/m

$\dot{E}_1(z) = 1\cdot e^{-j2\pi z}\,\hat{u}_x$ Vef/m

$\dot{H}_1(z) = 7,958\cdot e^{-j2\pi z}\,\hat{u}_y$ mAef/m

c) 

d) $k_{diel} \approx 2\pi\cdot 100\cdot 10^6\cdot\frac{3}{3\cdot 10^8}\cdot\left(1 - j\frac{0,06}{2\cdot 9}\right) = \beta - j\alpha$

$\alpha = 0{,}0067\pi$ Np/m; $\beta = 2\pi$ rad/m

$\eta_{diel} \approx \sqrt{\dfrac{\mu_0}{\epsilon'}} = 125{,}67\ \Omega$; $v = \dfrac{2\pi \cdot 10^8}{\beta} = 10^8$ m/s

e) $\dot{E}_1(z) = 1 \cdot e^{-j2\pi z} \cdot e^{-0{,}0067\pi z}\ \hat{u}_x$ Vef/m

$\dot{H}_1(z) = 7{,}958 \cdot e^{-j2\pi z}\ e^{-0{,}0067\pi z} \cdot \hat{u}_y$ mAef/m

f) $\vec{N}_{médio}(z) = 7{,}958 \cdot e^{-0{,}0134\pi z}\ \hat{u}_z$ mW/m²

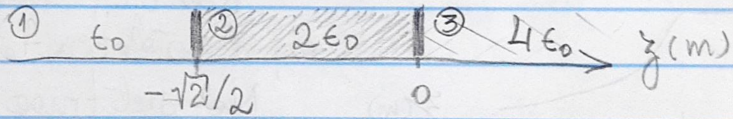
g) $P_{dis} = \vec{N}_{médio}(z=0) \cdot A = 0{,}00796$ W

23. a) $COE = \dfrac{|E|_{max}}{|E|_{min}} = 2 \Rightarrow |\rho| = \dfrac{1}{3} \Rightarrow \rho(z=0) = -\dfrac{1}{3}$

b) $\dfrac{\lambda_{ar}}{2} = 2$ m $\Rightarrow \lambda_{ar} = 4$ m ; $f = \dfrac{c}{\lambda} = 75$ MHz

c) $\eta_{diel} = \dfrac{\eta_0}{COE} = 188{,}5\ \Omega \Rightarrow \epsilon = COE^2 \cdot \epsilon_0 = 4\epsilon_0$

d)



$\rho_2(z=0) = \dfrac{\eta_3 - \eta_2}{\eta_3 + \eta_2} = \dfrac{1/2 - 1/\sqrt{2}}{1/2 + 1/\sqrt{2}} = \dfrac{\sqrt{2} - 2}{\sqrt{2} + 2} \cdot \dfrac{\sqrt{2} - 2}{\sqrt{2} - 2} = 2\sqrt{2} - 3$

$\beta_2 = 2\pi f \sqrt{\mu\epsilon} = 2\pi/4 \cdot \sqrt{2} = \pi\sqrt{2}/2$ rad/m

$\rho_2(z = -\sqrt{2}/2) = \rho_2(0) \cdot e^{-j2\frac{\pi\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}} = -\rho_2(0)$

$Z(z = -\sqrt{2}/2) = \eta_2 \dfrac{1 - \rho_2(0)}{1 + \rho_2(0)} = \dfrac{\eta_0}{\sqrt{2}} \cdot \dfrac{4 - 2\sqrt{2}}{2\sqrt{2} - 2} = \eta_0$

$\rho_1(z = -\dfrac{\sqrt{2}}{2}) = \dfrac{Z(-\sqrt{2}/2) - \eta_1}{Z(-\sqrt{2}/2) + \eta_1} = 0 \Rightarrow$ Sem reflexão no meio 1

24. a) $\delta = (\pi f \mu \sigma)^{-1/2} = 1{,}592$ mm ; $k = \dfrac{(1-j)}{\delta}$

$\eta_0 = 1/(\sigma e) \Rightarrow e = 26{,}525$ μm

$ke = (1-j)\dfrac{\sqrt{\pi f \mu_0 \sigma}}{\sigma \cdot \eta_0} = (1-j)\sqrt{\dfrac{\pi f \epsilon_0}{\sigma}} = (1-j)\,0{,}0167 \ll 1$

b) $\sigma = 100 \gg \omega\epsilon = 2\pi 10^9 \cdot 5\epsilon_0 = 0{,}278$ S/m

c) $\rho_{ar}(z=0) = \frac{\overbrace{0}^{\text{cond. perfeito}} - \eta_0}{0+\eta_0} = -1$; $\rho_{ar}(-\lambda/4) = -1 \cdot e^{-j2\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4}} = +1$

$Z(-\lambda/4) = \infty$

$\rho_{cond}(-\lambda/4) = \frac{\infty - \eta_{cond}}{\infty + \eta_{cond}} = 1$

$\rho_{cond}(-z_1) = \rho_{cond}(-\lambda/4)\cdot \exp\{-2\frac{1+j}{\delta}\cdot e\} = 0{,}967 \angle -1{,}91^\circ$

$Z(-z_1) = \frac{1+\rho_{cond}(-z_1)}{1-\rho_{cond}(-z_1)} \cdot \underbrace{(1+j)\frac{1}{\sigma\delta}}_{\eta_{cond}} = 377 \angle 0{,}01^\circ\ \Omega \approx \eta_0$

d) $\rho_{ar}(-z_1) = 0$; $P_- = |\rho|^2 P_+ = 0$ (0% reflet.)

c) ($f = 0{,}95$ GHz) $\delta = 1{,}633$ mm

$\lambda_0/4 = 75$ mm ; $\beta_{ar} = 19{,}897 \frac{rad}{m}$; $\rho_{ar}(-\lambda_0/4) = 1 \angle 9^\circ$

$Z(-\lambda_0/4) = 5 \cdot 10^3 \angle 90^\circ\ \Omega$; $\rho_{cond}(-\lambda_0/4) = +1$

$\rho_{cond}(-z_1) = 1 \cdot \exp\{-2\frac{(1+j)}{\delta}\cdot e\} = 0{,}968 \angle -1{,}86^\circ$

$Z(-z_1) = 377 \angle 0{,}01^\circ\ \Omega$

d) ($f = 0{,}95$ GHz) $\rho_{ar}(-z_1) = 0$; $P_- = 0\%$

25. a) $\beta_{ar} = 2\pi f/c = 20\pi \frac{rad}{m}$; $\beta_{jan} = 40\pi \frac{rad}{m}$

$\rho_{jan}(z=0) = \frac{\eta_0 - \eta_{jan}}{\eta_0 + \eta_{jan}} = \frac{1 - 1/2}{1+1/2} = \frac{1}{3}$

$\rho_{jan}(z=-0{,}1) = 1/3 \cdot \exp\{-j2\cdot 40\pi \cdot 0{,}1\} = 1/3$

$Z(-0{,}1) = 377\ \Omega$; $\rho_{ar}(-0{,}1) = 0$; $E_- = 0\%$

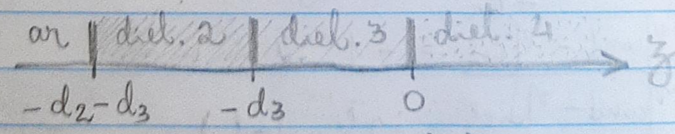
b) $\beta_{jan} = 20\pi$ rad/m ; $\rho_{jan}(-0{,}1) = 1/3$

$Z(-0{,}1) = 377\ \Omega$; $\rho_{ar}(-0{,}1) = 0$; $E_- = 0\%$

c) $\beta_{jan} = 4\pi$ rad/m ; $\rho_{jan}(-0{,}1) = 1/3 \angle -144^\circ$

$Z(-0{,}1) = 110{,}948 \angle -23{,}79^\circ\ \Omega$; $\rho_{ar}(-0{,}1) = 0{,}58 \angle 14{,}57^\circ$

$E_- = 58\%$; $P_- = 33{,}72\%$

26. 

$\rho_3(0) = \frac{\eta_4 - \eta_3}{\eta_4 + \eta_3} \rightarrow \rho_3(-d_3) = \rho_3(0)\cdot e^{-j2\cdot\frac{2\pi}{\lambda_3}\cdot\frac{\lambda_3}{4}} = -\rho_3(0)$

$Z(-d_3) = \eta_3^2/\eta_4 \rightarrow \rho_2(-d_2-d_3) = -\rho_2(-d_3) = -\frac{Z(-d_3)-\eta_2}{Z(-d_3)+\eta_2}$

$$Z(-d_2-d_3) = \frac{\eta_2^2}{Z(-d_3)} = \frac{\eta_2^2}{\eta_3^2}\cdot\eta_4$$

Para casamento perfeito, ρ_1 deve ser zero:

$$\rho_1(-d_2-d_3) = \frac{Z(-d_2-d_3)-\eta_1}{Z(-d_2-d_3)+\eta_4} = 0$$

Ou, $\left(\frac{\eta_2}{\eta_3}\right)^2\cdot\eta_4 = \eta_1 \to \frac{\eta_2}{\eta_3} = \left(\frac{\eta_1}{\eta_4}\right)^{1/2}$, Q.E.D.

■ Para frequência 10% abaixo, $\beta_i = 0,9\cdot\frac{2\pi}{\lambda_{i0}} = \frac{1,8\pi}{\lambda_{i0}}$ para $i \in \{1,2,3,4\}$ e λ_{i0} do casamento.

$\rho_3(0) = -0,172 \to \rho_3(-d_3) = 0,172\ \angle 18°$

$Z_3(-d_3) = 185,092\ \angle 6,24°\ \Omega \to \rho_2(-d_3) = 0,188\ \angle 163,764°$

$\rho_2(-d_2-d_3) = 0,188\ \angle 1,764° \to Z(-d_2-d_3) = 390,3\ \angle 0,69°\ \Omega$

E, finalmente, $\rho_1(-d_2-d_3) = 0,01835\ \angle 19,13°$

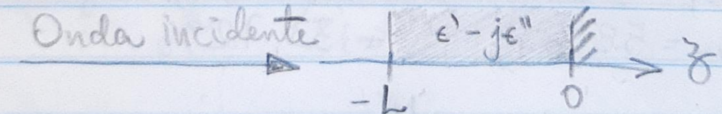
■ Na figura (b), $\rho_2(0) = \frac{\eta_4-\eta_2}{\eta_4+\eta_2} = -1/3$

$\rho_2(-\lambda_{20}/4) = 1/3\ \angle 18° \to Z(-\lambda_{20}/4) = 360,525\ \angle 13,05°\ \Omega$

$\rho_1(-\lambda_{20}/4) = 0,11653\ \angle 101,2°$

Ou seja, o caso em (a) é mais resistente a variações de frequência (reduz as reflexões numa banda maior).

27.



$$\rho_{diel}(z=0) = \frac{\eta_{cond}-\eta_{diel}}{\eta_{cond}-\eta_{diel}} = -1$$

$$\rho_{diel}(z=-L) = \rho_{diel}(0)\cdot e^{j2k_{diel}(-L)} = -e^{-j2k_{diel}\cdot L}$$

$$Z(-L) = \eta_{diel}\cdot\frac{1+\rho_{diel}(-L)}{1-\rho_{diel}(-L)} \quad \begin{cases} k_{diel} \approx \omega\sqrt{\mu\epsilon'}\,(1-j\epsilon''/2\epsilon') \\ \eta_{diel} \approx \sqrt{\mu/\epsilon'} \end{cases}$$

$k_{diel} = 2\pi 3.10^9 \cdot \frac{2}{3.10^8}\left(1 - j\frac{0{,}01}{4}\right) = 10\pi(4 - j0{,}01)\ \frac{rad}{m}$

$\eta_{diel} = \eta_0/2 = 188{,}5\ \Omega$

$\rho_{diel}(-L) = 0{,}9922\angle 0°$ e $Z(-L) = 48\,k\Omega \approx \infty$

(Poderia, por exemplo, ser notado que $L = \lambda_{diel}/4$, e, consequentemente, o efeito do condutor aparece na interface dielétrico/vácuo como um aberto $Z: 0 \xrightarrow{\lambda/4} \infty$)

28\. a) $\eta_3 = \sqrt{\frac{\mu}{\epsilon_3}} = \frac{\eta_0}{\sqrt{9}} = 125{,}67\,\Omega$; $\eta_1 = \eta_0$

$k_3 = 2\pi.10^9\sqrt{\mu\epsilon_3} = 2\pi\, {}^{10}/_3 \cdot \sqrt{9} = 20\pi$ rad/m

$k_1 = 2\pi.10^9/c = 20\pi/3$ rad/m

b) Para $L = \lambda_2/4$: $\rho_2(-L) = \rho_2(0)\,e^{-j\pi} = -\rho_2(0)$

$\rightarrow Z(-L) = \eta_2 \cdot \frac{1-\rho_2(0)}{1+\rho_2(0)} = \eta_2^2/Z(0) = \eta_2^2/\eta_3$

Fazendo máxima incidência de potência (sem reflexão): $Z(-L) = \eta_1 \rightarrow \eta_2 = \sqrt{\eta_1 \cdot \eta_3} = \sqrt{\frac{\eta_0^2}{3}} = \frac{\eta_0}{\sqrt{3}}$

Como $\eta_2 = \eta_0/\sqrt{\epsilon_2}$, sai que $\epsilon_2 = \sqrt{\epsilon_1 \cdot \epsilon_3} = 3$

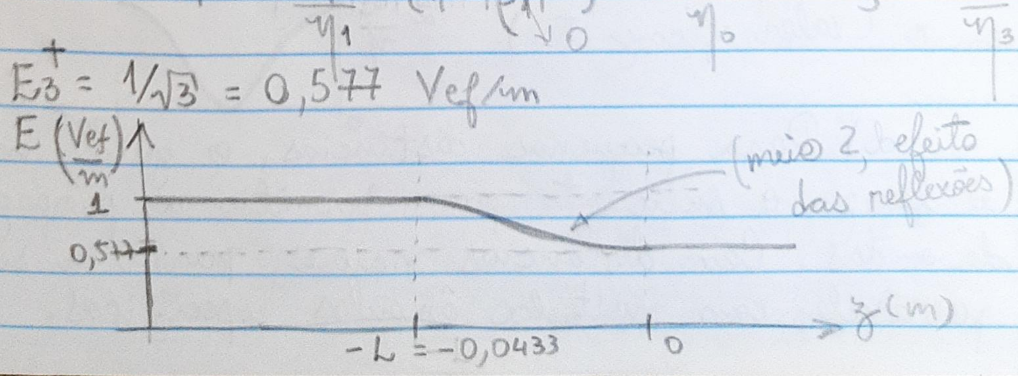
$L = (\underbrace{v_2/f}_{\lambda_2})/4 = \frac{c}{\sqrt{\epsilon_2}\,f} \cdot \frac{1}{4} = 4{,}33$ cm

c) $\rho_3(0) = 0$ $\quad \rho_2(0) = \frac{\eta_3 - \eta_2}{\eta_3 + \eta_2} = -0{,}268$

$\rho_2(-L) = +0{,}268 \rightarrow Z(-L) = \eta_0 \rightarrow \rho_1(-L) = 0$

d) $\vec{N}_1 = \frac{|E_1^+|^2}{\eta_1}(1 - |\rho_1|^2) = \frac{1}{\eta_0} = \vec{N}_3 = \frac{|E_3^+|^2}{\eta_3}(1 - |\rho_3|^2)$, com $\rho_1 \rightarrow 0$ e $\rho_3 \rightarrow 0$

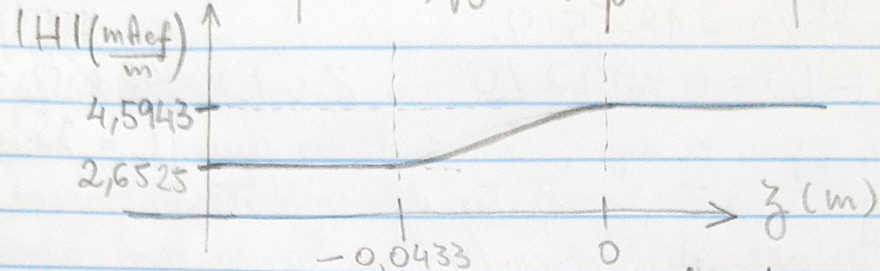
$\rightarrow E_3^+ = 1/\sqrt{3} = 0{,}577$ Vef/m



(14)

e) $\dot{\vec{H}}_1^{+} = 1/\eta_0 \cdot e^{-jk_1 z}\,\hat{u}_y \rightarrow |H_1| = 1/\eta_0 = 2{,}65$ mAef/m

$|H_3| = |\dot{E}_3|/\eta_3 = \frac{1}{\sqrt{3}} \cdot 3/\eta_0 = \sqrt{3}/\eta_0 = 4{,}59$ mAef/m



f) $\rho_2(-L) = \rho_2(0) \cdot e^{-j2 \cdot k_2 \cdot L}$

Com $k_2 = 2\pi \cdot 2 \cdot 10^9 \cdot \sqrt{3}/c$ e $L = c/(4\sqrt{3} \cdot 10^9)$

ou seja, $\rho_2(-L) = \rho_2(0)$ (pois agora $L = \lambda_2/2$).

$Z(-L) = \eta_3$ e $\rho_1(-L) = -0{,}5$

29. a) Transversal magnética (TM)

b) Pelas condições de contorno, impõe-se que

$E_x(x,z) = -j2\dot{E}_+ \cos\theta \,\text{sen}(kz\cos\theta)\, e^{-jkx\,\text{sen}\theta}$

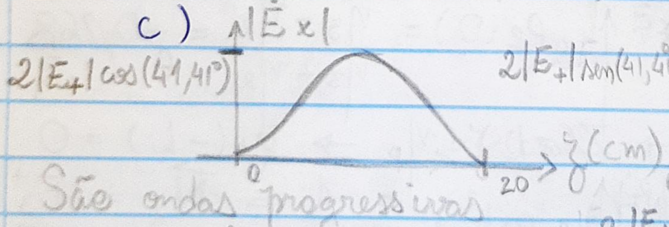
seja zero nas paredes do guia, ou seja

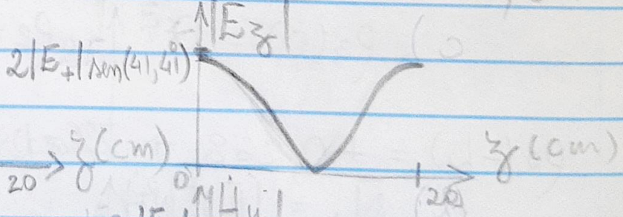
$k\,a\cos\theta = n\pi \rightarrow \theta = \cos^{-1}(n\lambda/(2a))$ $\left\{ n \in \mathbb{Z} \text{ e } \lambda = c/f \right.$

- $a = 10$ cm : $\theta \nexists$
- $a = 15$ cm : $\theta = 0°$
- $a = 20$ cm : $\theta = 41{,}41°$

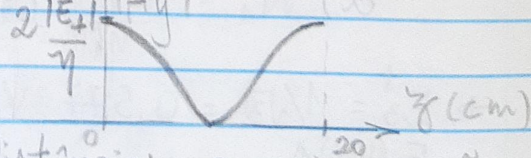
$\bullet\ a = 50$ cm : $\theta = \begin{cases} 72{,}54° \\ 53{,}13° \\ 25{,}84° \end{cases}$

c)





São ondas progressivas em x (valor eficaz cte.)



d) Para pequenas distâncias, o guia não é largo o suficiente para permitir a propagação de ondas. Com distâncias maiores, passa a ser possível, com múltiplos ângulos específicos.

30. $\dot{\vec{E}}(r, \theta = 1km, \pi/2) = j\eta_0 k \frac{h I_0}{4\pi} sen\theta \left[\frac{e^{-jkr_1}}{r_1} + \frac{e^{-jkr_2}}{r_2}\right]\hat{u}_\theta$

$$\underbrace{}_{r_1 \approx r_2 \approx 10^3} \approx j\eta_0 k \frac{h I_0 e^{-jk10^3}}{4\pi 10^3}\left[e^{-jk0{,}25} + e^{+jk0{,}25}\right]\hat{u}_\theta$$

$$= j\eta_0 k \frac{h I_0}{4\pi 10^3} e^{-jk10^3} \cos(k \cdot 0{,}25)\,\hat{u}_\theta$$

Como $k \cdot 0{,}25 = \frac{2\pi}{\lambda} \cdot \frac{1}{4} = \frac{\pi}{2} \cdot \frac{f}{c} = \frac{\pi}{2} \rightarrow \dot{\vec{E}} = \dot{\vec{H}} = 0$

31. $\dot{\vec{E}}^{+}_{xy1}(x,0) + \dot{\vec{E}}^{-}_{xy1}(x,0) = \dot{\vec{E}}^{+}_{xy2}(x,0) \rightarrow$

$$[\dot{E}^{+}_{y1}\hat{u}_y + \dot{E}^{+}_{\xi 1}\hat{u}_\xi]e^{-jk_1\xi} + [\dot{E}^{-}_{y1}\hat{u}_y + \dot{E}^{-}_{\xi 1}\hat{u}_{\xi'}]e^{-jk_1\xi'}$$
$$= [\dot{E}^{t}_{y2}\hat{u}_y + \dot{E}^{t}_{\xi 2}\hat{u}_{\xi''}]e^{-jk_2\xi''}$$

Da figura, $\dot{E}_y = 0$:

- $\dot{E}^{+}_{\xi 1}\cos(\theta)e^{-jk_1 x\, sen\theta} + \dot{E}^{-}_{\xi 1}\cos(\theta')e^{-jk_1 x\, sen\theta'} = \dot{E}^{t}_{\xi 2}\cos(\theta'')e^{-jk_2 x\, sen\theta''}$

Analogamente:

- $\dot{H}^{+}_{y1} e^{-jk_1 x\, sen\theta} + \dot{H}^{-}_{y1} e^{-jk_1 x\, sen\theta'} = \dot{H}^{t}_{y2} e^{-jk_2 x\, sen\theta''}$

32. a) $\hat{u}_x = \hat{u}_\xi = \hat{u}_x\cos\theta - \hat{u}_z sen\theta$

As componentes tangenciais (x, y) devem ser contínuas:

- $E_{01+}\cos\theta\, e^{-jk_0 x\, sen\theta} + E_{01-}\cos\theta'\, e^{-jk_0 x\, sen\theta'} =$
$$E_{02+}\cos\theta''\, e^{-jk_2 x\, sen\theta''}$$

- $\frac{E_{01+}}{\eta_0} e^{-jk_0 x\, sen\theta} + \frac{E_{01-}}{\eta_0} e^{-jk_0 x\, sen\theta'} = \frac{E_{02+}}{\eta_2} e^{-jk_2 x\, sen\theta''}$

b) Como as variações em x devem ser iguais, para que continuem válidas as equações:

$\beta_0 sen\theta \overset{(1)}{=} \beta_0 sen\theta' \overset{(2)}{=} \beta_2 sen\theta''$ ($\beta = Re\{k\}$, para análise de fase)

(1) $\Rightarrow sen\theta = sen\theta' \rightarrow \theta' = \theta$

(2) $\Rightarrow sen\theta = sen\theta'' \rightarrow \frac{sen\theta}{v_1} = \frac{sen\theta''}{v_2} \rightarrow \frac{sen\theta_2}{sen\theta} = \frac{n_1}{n_2} = \sqrt{\frac{\epsilon_1}{\epsilon_2}}$

$$\theta'' = arcsen\left(\frac{k_0}{k_2} sen\theta\right)$$

33.